Andy Zhang

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Education

University of Waterloo — Waterloo, ON

Bachelor's degree, Computer Science, Relevant Coursework: CS145, CS146, CS136L

- Recipient of the National Math Scholarship, valued at **$15,000** awarded to 15 students

Technical Skills

Languages: Python, Java, C/C++, SQL, R, JavaScript, TypeScript, HTML/CSS
Developer Tools: Git, Docker, Kubernetes, GitLab CI/CD, Azure, AWS, Grafana, Artifactory, Vercel, Modal
Libraries/Frameworks: React, FastAPI, Flask, Swift, Shiny, PyTorch, OpenCV, Pytest, DuckDB, Pandas, NumPy

Experience

Statistics Canada — May 2026 – Aug 2026

Software Engineering Intern — *Ottawa, ON*

- Engineered containerized internal web applications **enabling 100+ employees** to process, validate, and manage survey and census datasets more efficiently through Helm charts, CI/CD pipelines, and Kubernetes workflows
- **Implemented Azure Blob Storage for 5 internal Python/Shiny applications** by designing reusable storage configuration, resolving identity/permission issues, and standardizing file access across dev/test deployments
- Optimized XML to Parquet **conversion speed by 6.9x** by improving DuckDB operations, selecting efficient batch sizes, and reducing repeated read/write operations during large file conversions, reducing runtime by 20 hours
- Automated deployments for web applications by configuring GitLab CI pipelines to trigger Kubernetes releases

Nokia — July 2024 – Aug 2024

Software Engineering Intern — *Remote*

- Developed automated Python performance tests for the Network Services Platform team, validating network workflows under repeated load, improving test coverage, execution consistency, and regression stability
- Built **scalable performance testing environments** using Locust and Kubernetes, simulating network activity and monitoring bottlenecks through Grafana and Prometheus dashboards to **reduce analysis runtime by 75%**
- Created real-time Python data collection and reporting scripts to parse network performance metrics from test runs, generate summary reports, and streamline troubleshooting workflows, **improving analysis efficiency by 100%**

Projects

VR Memory Reconstruction App | *PyTorch, Gaussian Splatting, Unity, Swift, React, FastAPI, Vercel, Modal*

- Built a **full-stack platform** that turns user uploaded videos from a React web app or Swift app into explorable VR worlds, using a FastAPI backend to run 3D Gaussian Splat reconstruction and Unity OpenXR for VR playback
- Deployed a React/Vite platform that turns uploaded videos into VR scenes, using **Modal for GPU backend** processing, cloud storage, and horizontal scaling for large reconstructions and to support scalable upload workflows
- Orchestrated a reconstruction pipeline that extracts video frames, runs COLMAP sparse reconstruction and FastGS training, importing the generated PLY assets into Unity as reusable VR scene prefabs for interactive playback

Catan Engine - AlphaHex | *PyTorch, FastAPI, React, Vite, MCTS, Neural Networks*

- Achieved a **top 500 online rank with a 70% win rate** in 1v1 Catan by building an **MCTS engine** that evaluates moves through rollout search, board heuristics, and long-term resource/expansion strategy modeling
- Improved engine decisions through **self-play training** by developing an **MCTS-NN policy pipeline** with replay buffers, checkpointed models, and leaderboard evaluation to iteratively tune strategy against prior engine versions

AI Legal Research Assistant | *Flask, Tesseract OCR, Pinecone, Google Cloud Services*

- Designed a Chrome extension legal assistant that processes PDFs and images with Tesseract OCR, extracts key legal content, and uses a Flask backend with **GPT-4o-mini** to generate concise summaries and explanations
- Added a **RAG-style document memory system** with Pinecone vector search, embedding uploaded documents into searchable chunks and retrieving the most relevant passages for user-specific and context aware legal help

Poker Bluff Detection Engine | *Python, OpenCV, MediaPipe, DeepFace*

- Constructed a **real-time poker bluff detection system** using OpenCV, MediaPipe, and DeepFace to analyze facial cues such as blinking, lip biting, head movement, and emotion changes to estimate player behavioral patterns
- Formulated an **algorithm to calculate bluff scores** by combining behavioral indicators and weighting facial cues